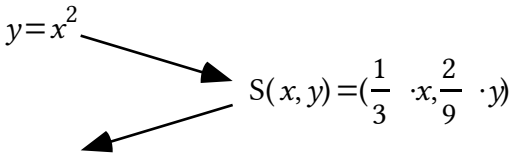
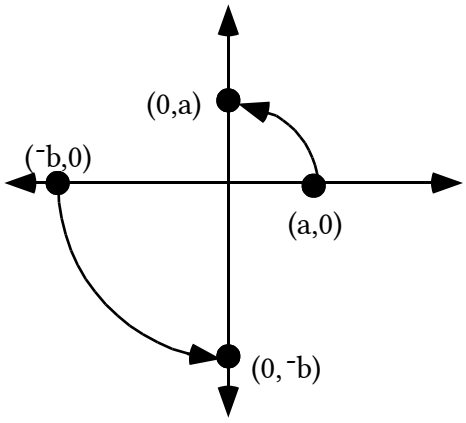
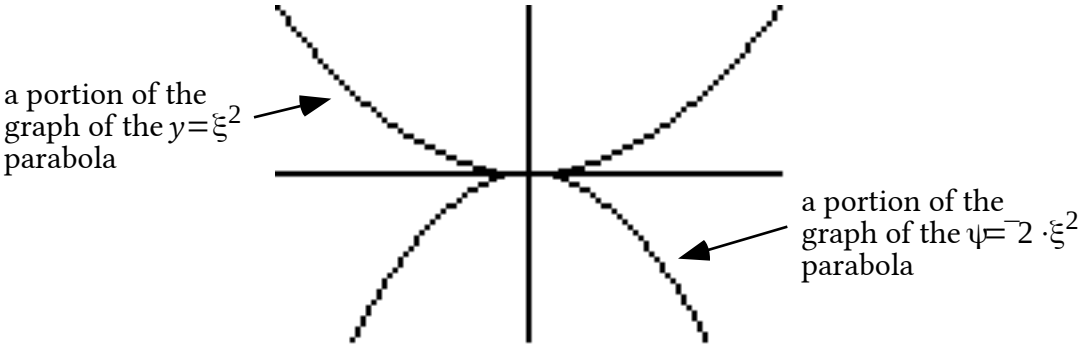

$y=x^2$
$S(x,y)=(-1\cdot x,\frac{3}{4}\cdot y)$

$$y = 16$$


$y=x^2$
$S(x,y)=(\frac{1}{3}\cdot x,\frac{2}{9}\cdot y)$


(0,a)
(-b,0)
(a,0)
(0,-b)

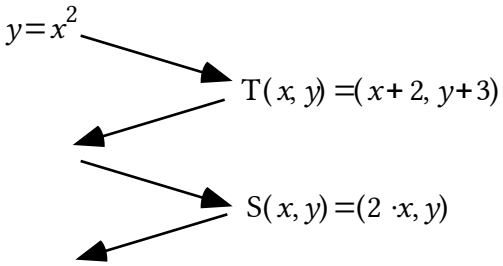

a portion of the graph of the $y=\xi^2$ parabola
a portion of the graph of the $\psi=-2\cdot\xi^2$ parabola

$x^2 + 6x + \underset{B}{____} = 8 + \underset{C}{____}$

$(x + \underset{A}{____})^2 = \underset{D}{____}$

$x^2 + {}^{-}10x + \underset{B}{____} = {}^{-}11 + \underset{C}{____}$

$(x + \underset{A}{____})^2 = \underset{D}{____}$


$y=x^2$
$T(x, y) = (x+2, y+3)$
$S(x, y) = (2 \cdot x, y)$

$x^2 + 12x + \underset{B}{____} = -11 + \underset{C}{____}$

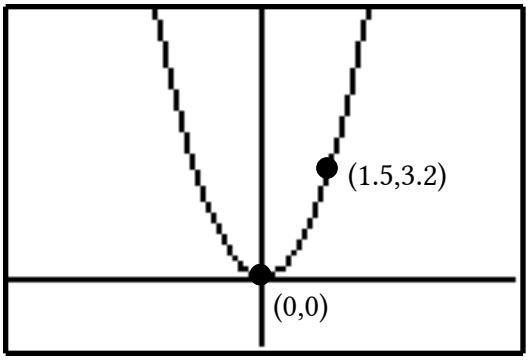
$(x + \underset{A}{____})^2 = \underset{D}{____}$

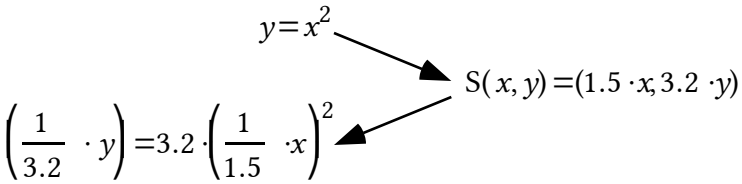
$x^2 + 7x + \underset{B}{____} = -13 + \underset{C}{____}$

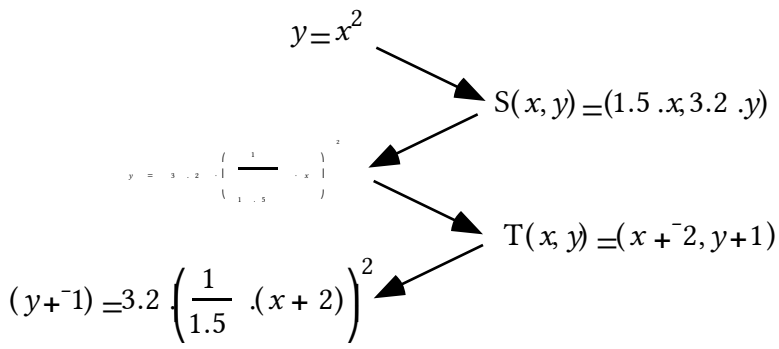
$(x + \underset{A}{____})^2 = \underset{D}{____}$

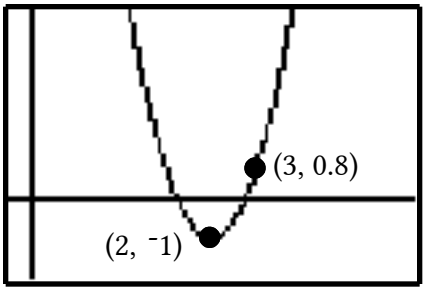
$x^2 + \sqrt{5}\,x + \underset{B}{____} = -1 + \underset{C}{____}$

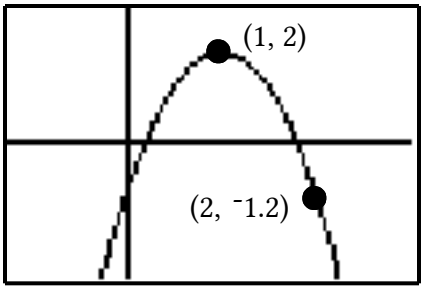
$(x + \underset{A}{____})^2 = \underset{D}{____}$


(1.5,3.2)
(0,0)


$y = x^2$
$S(x, y) = (1.5 \cdot x, 3.2 \cdot y)$
$\left(\frac{1}{3.2} \cdot y\right) = 3.2 \cdot \left(\frac{1}{1.5} \cdot x\right)^2$


$y = x^2$
$S(x, y) = (1.5 \cdot x, 3.2 \cdot y)$
$y = 3.2 \cdot \left(\frac{1}{1.5} \cdot x \right)^2$
$T(x, y) = (x + 2, y + 1)$
$(y + 1) = 3.2 \cdot \left(\frac{1}{1.5} \cdot (x + 2) \right)^2$


(3, 0.8)
(2, -1)


(1, 2)
(2, -1.2)

$$x^2 = 5$$

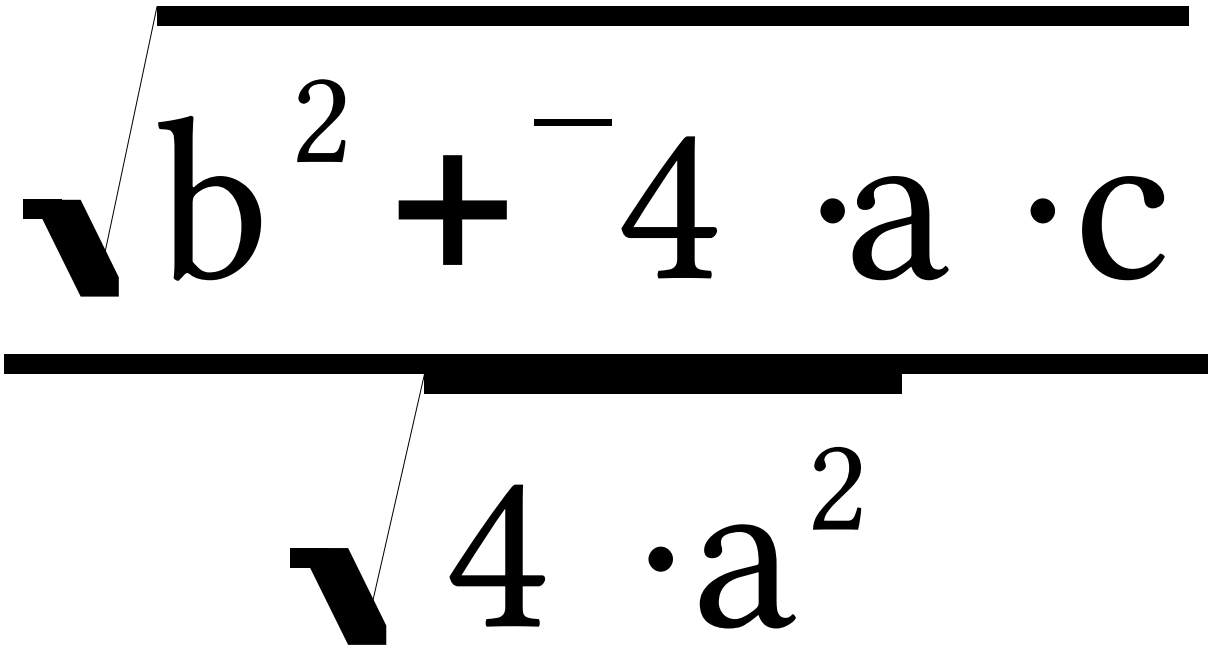
$$\left(x+\frac{\sqrt{11}}{2}\right)^2=5$$

$x^2 + \sqrt{11}\,x + \underset{\text{B}}{____} = \frac{5}{4} + \underset{\text{C}}{____}$

$(x + \underset{\text{A}}{____})^2 = \underset{\text{D}}{____}$

$$x^2 + \frac{b}{a} \cdot x + \underbrace{\frac{b^2}{4 \cdot a^2}}_{B} = \underbrace{-\frac{c}{a}}_{} + \underbrace{\frac{b^2}{4 \cdot a^2}}_{C}$$

$$\underbrace{\left(x + \frac{1}{2} \cdot \frac{b}{a}\right)^2}_{A} = \underbrace{\frac{-4 \cdot a \cdot c + b^2}{4 \cdot a^2}}_{D}$$



$$\frac{-\sqrt{b^2 + -4 \cdot a \cdot c}}{\sqrt{4 \cdot a^2}}$$

$$x^2 = \frac{b^2 + {}^{-}4 \cdot a \cdot c}{4 \cdot a^2}$$

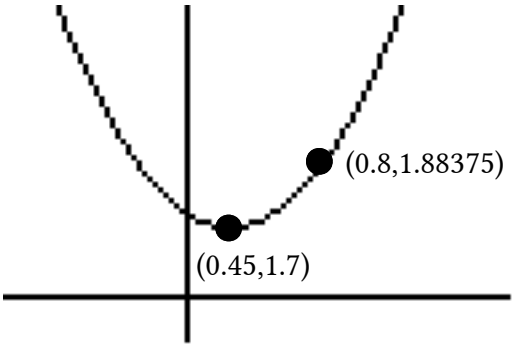
$$\left(x+\frac{b}{2\cdot a}\right)^2=\frac{b^2+^-4\cdot a\cdot c}{4\cdot a^2}$$

$$x + \frac{\urcorner b}{2 \cdot a}$$

$$\frac{-b}{2 \cdot a} + \frac{\sqrt{b^2 + -4 \cdot a \cdot c}}{\sqrt{4 \cdot a^2}}$$

$$\frac{-b}{2 \cdot a} + \frac{-\sqrt{b^2 + -4 \cdot a \cdot c}}{\sqrt{4 \cdot a^2}}$$

$$\frac{-b}{2 \cdot a} + \frac{-\sqrt{b^2 + -4 \cdot a \cdot c}}{\sqrt{4 \cdot a^2}}$$


(0.8,1.88375)
(0.45,1.7)

$$\frac{-b}{2 \cdot a} + \frac{\sqrt{b^2 + -4 \cdot a \cdot c}}{2 \cdot a}$$

$$\frac{-b}{2 \cdot a} + \frac{-\sqrt{b^2 + -4 \cdot a \cdot c}}{2 \cdot a}$$

$$\frac{-b + \sqrt{b^2 + -4 \cdot a \cdot c}}{2 \cdot a}$$

$$\frac{-b + \sqrt{b^2 + -4 \cdot a \cdot c}}{2 \cdot a}$$

$$\frac{-b}{2 \cdot a} + \frac{\sqrt{b^2 + -4 \cdot a \cdot c}}{-2 \cdot a}$$

$$\frac{-b}{2 \cdot a} + \frac{-\sqrt{b^2 + -4 \cdot a \cdot c}}{-2 \cdot a}$$

$$\frac{-b}{2a}+\frac{-1\cdot\sqrt{b^2+-4\cdot a\cdot c}}{-1\cdot-2\cdot a}=\frac{-b}{2\cdot a}+\frac{-\sqrt{b^2+-4\cdot a\cdot c}}{2\cdot a}=\frac{-b+-\sqrt{b^2+-4\cdot a\cdot c}}{2\cdot a}$$

$$\frac{-b}{2 \cdot a} + \frac{-1 \cdot -\sqrt{b^2 + -4 \cdot a \cdot c}}{-1 \cdot -2 \cdot a} = \frac{-b}{2 \cdot a} + \frac{-1 \cdot -1\sqrt{b^2 + -4 \cdot a \cdot c}}{2 \cdot a} = \frac{-b + \sqrt{b^2 + -4 \cdot a \cdot c}}{2 \cdot a}$$

$$\left[\begin{array}{cc|cc} 5 & 3 & 1 & 0 \\ 4 & 3 & 0 & 1 \end{array}\right]$$

A_1
B_1

$$\left[\begin{array}{cc|cc} 1 & \frac{3}{5} & \frac{1}{5} & 0 \\ 4 & 3 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{3}{5} & \frac{1}{5} & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} & 1 \end{array}\right]$$

A_3 =

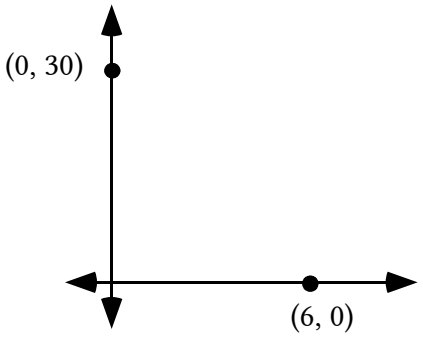
B_3 =

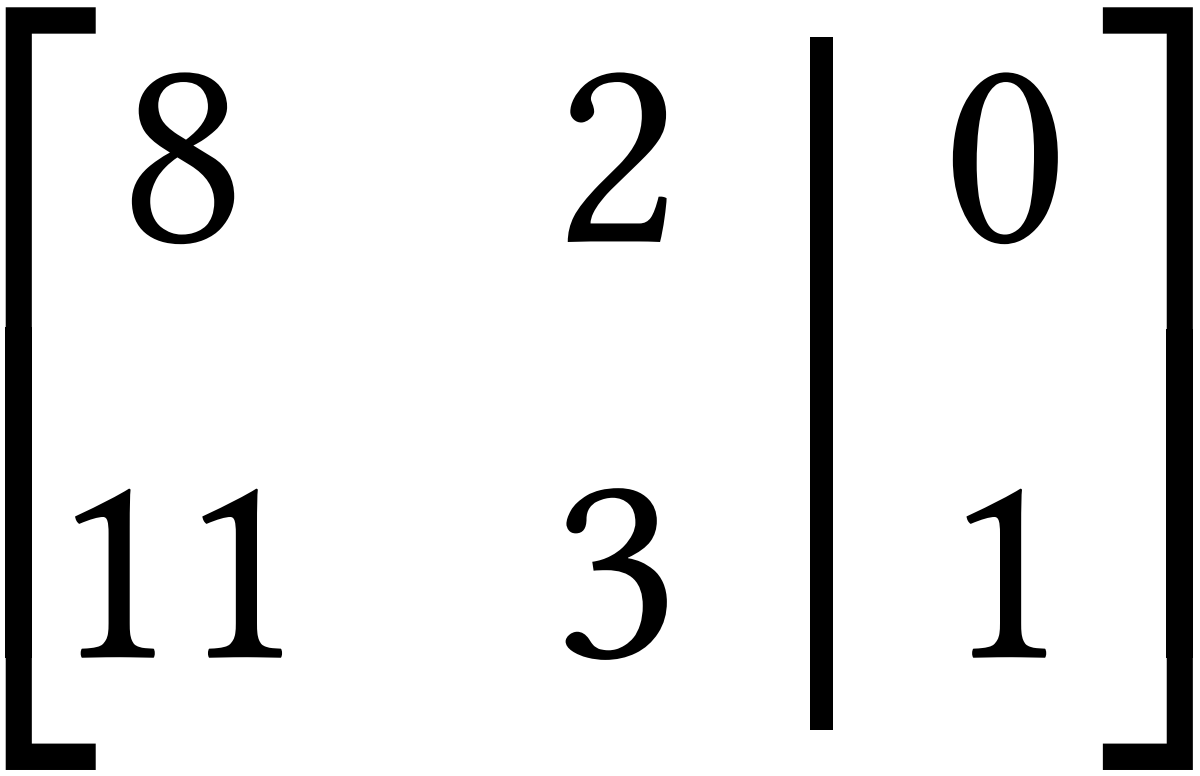
$$\left[\begin{array}{cc|cc} 1 & \frac{3}{5} & \frac{1}{5} & 0 \\ 0 & 1 & -\frac{4}{3} & \frac{5}{3} \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -\frac{4}{3} & \frac{5}{3} \end{array}\right]$$

25

25


(0, 30)
(6, 0)


8 2 0
11 3 1

$$\left[\begin{array}{ccc|ccc} 4 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 9 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \begin{matrix} A_1 \\ B_1 \\ C_1 \end{matrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{-1}{2} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 9 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} A_2 = \\ B_2 = \\ C_2 = \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{-1}{2} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ 0 & \frac{3}{2} & \frac{3}{4} & \frac{-1}{4} & 1 & 0 \\ 0 & \frac{15}{2} & \frac{-5}{4} & \frac{-9}{4} & 0 & 1 \end{array}\right] \begin{array}{l} A_3 = \\ B_3 = \\ C_3 = \end{array}$$

A_4 =

B_4 =

C_4 =

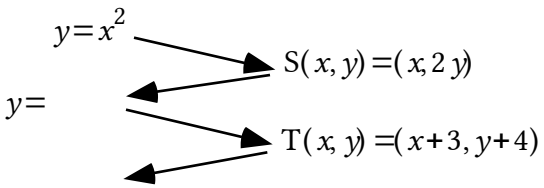
A_5 =

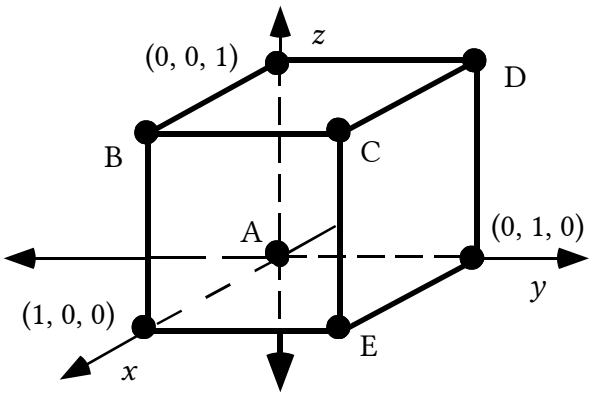
B_5 =

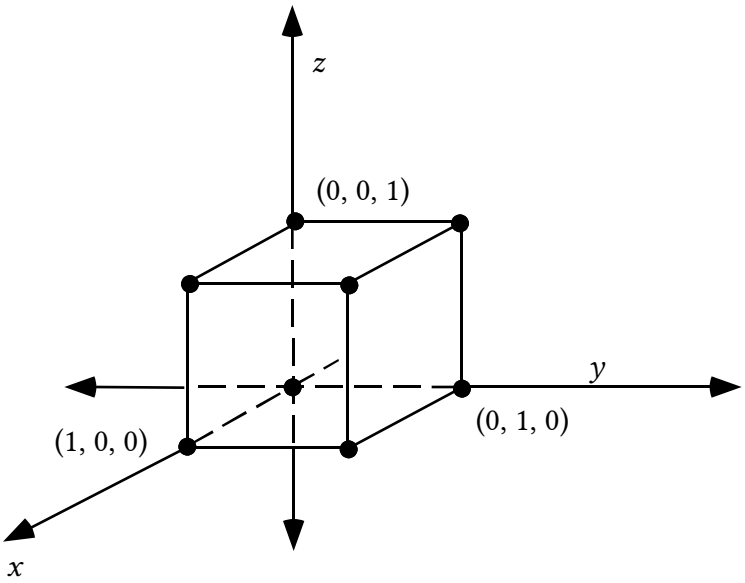
C_5 =

$$\left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} & \frac{-1}{6} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & \frac{1}{5} & 1 & \frac{-1}{5} \end{array}\right] \begin{array}{l} A_6 = \\ B_6 = \\ C_6 = \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{15} & \frac{-1}{6} & \frac{1}{10} \\ 0 & 1 & 0 & \frac{-4}{15} & \frac{1}{6} & \frac{1}{10} \\ 0 & 0 & 1 & \frac{1}{5} & 1 & \frac{-1}{5} \end{array}\right] \begin{array}{l} A_7 = \\ B_7 = \\ C_7 = \end{array}$$

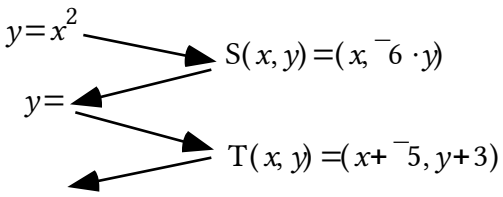

$y=x^2$
$S(x, y)=(x, 2y)$
$y=$
$T(x, y)=(x+3, y+4)$

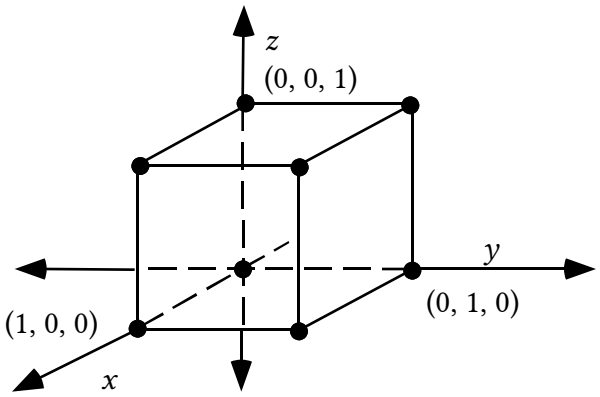

z
(0, 0, 1)
D
B
C
A
(0, 1, 0)
y
(1, 0, 0)
E
x


z
(0, 0, 1)
y
(0, 1, 0)
(1, 0, 0)
x

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \cdot \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{e \cdot i - f \cdot h}{a \cdot (e \cdot i - f \cdot h) + -b \cdot (d \cdot i - g \cdot f) + c \cdot (d \cdot h - g \cdot e)} \\ 0 & 1 & 0 & \frac{-1 \cdot (d \cdot i - f \cdot g)}{a \cdot (e \cdot i - f \cdot h) + -b \cdot (d \cdot i - g \cdot f) + c \cdot (d \cdot h - g \cdot e)} \\ 0 & 0 & 1 & \frac{(d \cdot h - g \cdot e)}{a \cdot (e \cdot i - f \cdot h) + -b \cdot (d \cdot i - g \cdot f) + c \cdot (d \cdot h - g \cdot e)} \end{array}\right]$$


$y=x^2$
$S(x,y)=(x,\bar{6}\cdot y)$
$y=$
$T(x,y)=(x+\bar{5},y+3)$


z
(0, 0, 1)
y
(0, 1, 0)
(1, 0, 0)
x

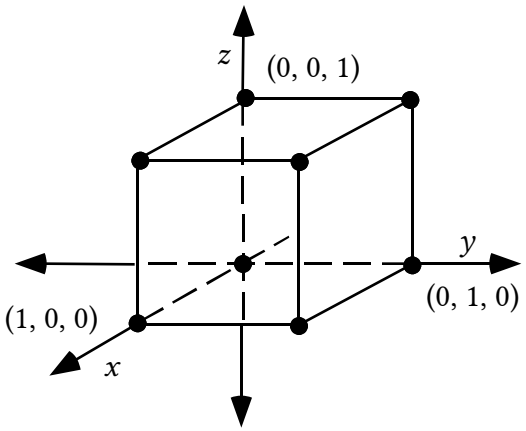
	pre-image	image
perimeter		
surface area		
volume		

$y = 2 \cdot x^2 + 10 \cdot x + {}^{-}3$

$y = 2 \cdot (x^2 + ___ \cdot x) + {}^{-}3$

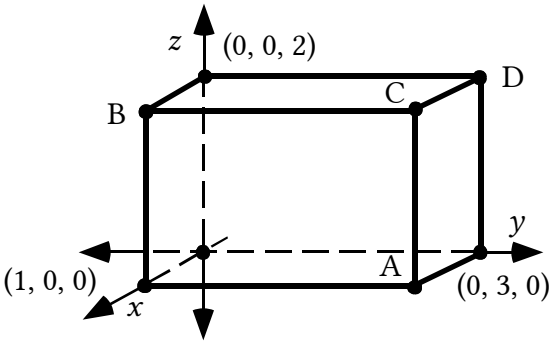
$y = 2 \cdot (x^2 + 5 \cdot x + \frac{25}{4}) + {}^{-}3 + ___$

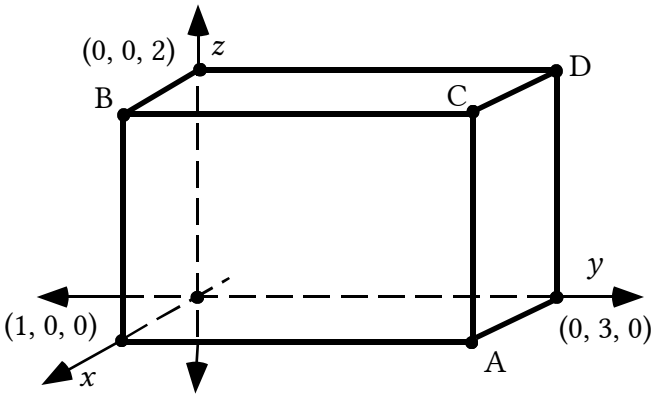
$y = 2 \cdot (x + ___)^2 + ___$

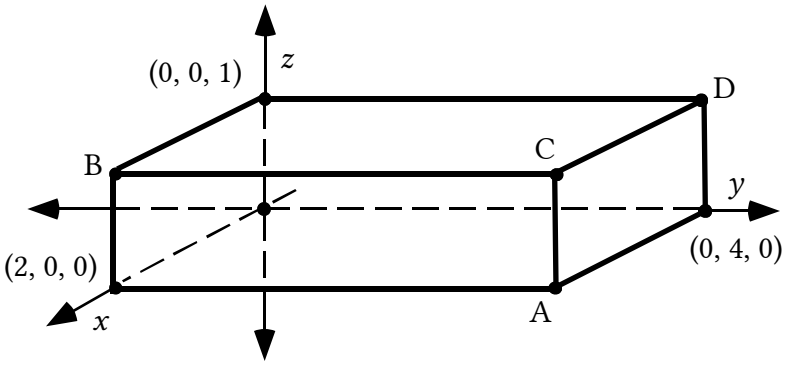

z
(0, 0, 1)
y
(0, 1, 0)
(1, 0, 0)
x

magnitude of size transformation	perimeter	surface area	volume
1			
2			
3			
4			

magnitude of size transformation	perimeter	surface area	volume
1	12	6	1
2	24	24	8
3	36	54	27
4	48	96	64


z
(0, 0, 2)
D
C
B
y
A
(1, 0, 0)
(0, 3, 0)
x

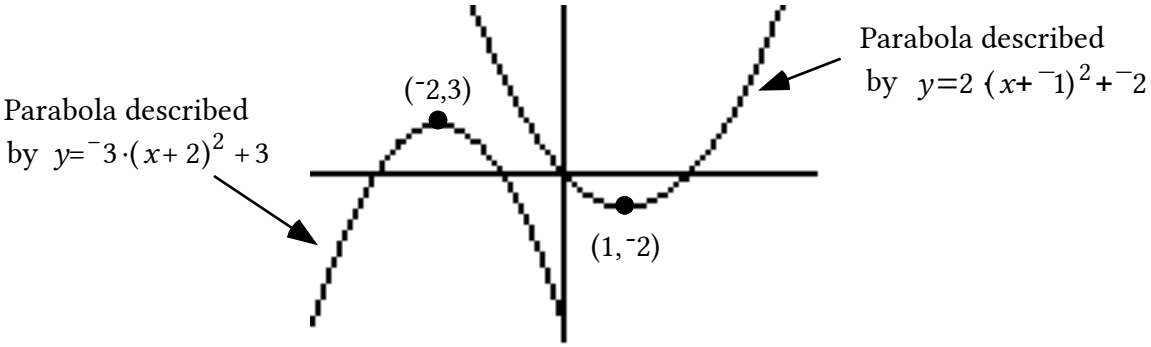

(0, 0, 2)
z
D
B
C
y
(1, 0, 0)
(0, 3, 0)
x
A

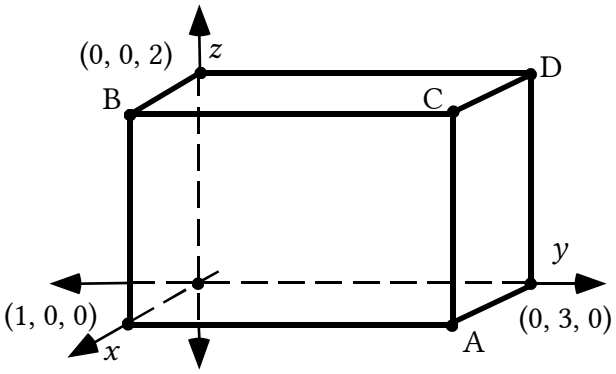

(0, 0, 1)
z
D
B
C
y
(0, 4, 0)
(2, 0, 0)
x
A

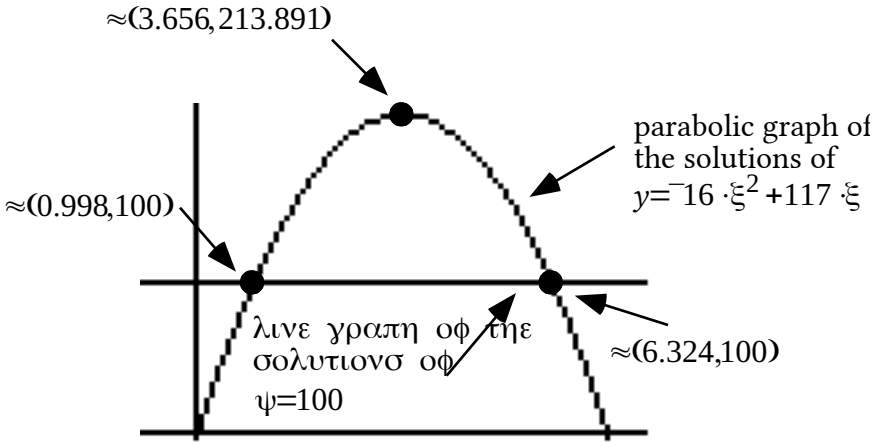
$$\bar{1}6 \cdot x^2 + 117 \cdot x = 100$$

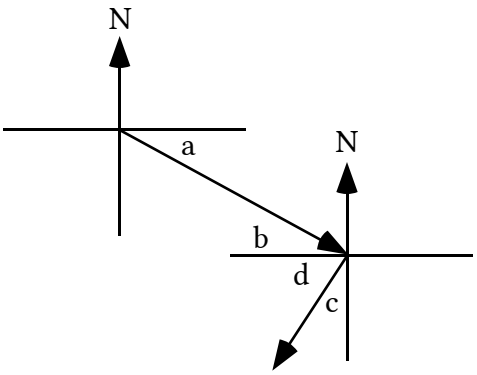
$$\bar{}16 \cdot x^2 + 117 \cdot x + \bar{}100 = 0$$

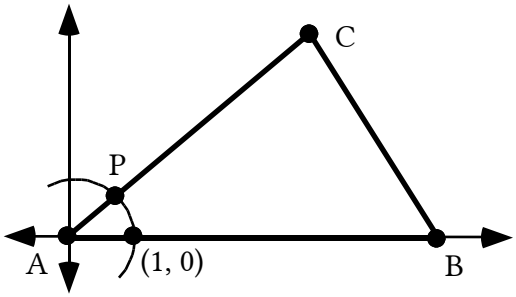
$$y = -16 \cdot x^2 + 117 \cdot x$$

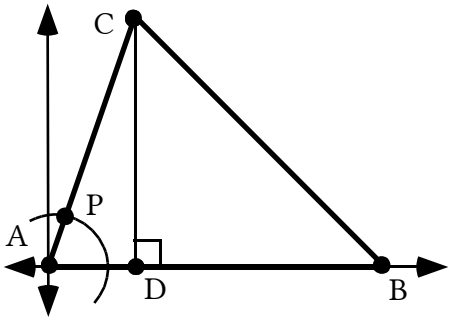

Parabola described by $y=-3\cdot(x+2)^2+3$
(-2,3)
(1,-2)
Parabola described by $y=2\cdot(x+-1)^2+-2$

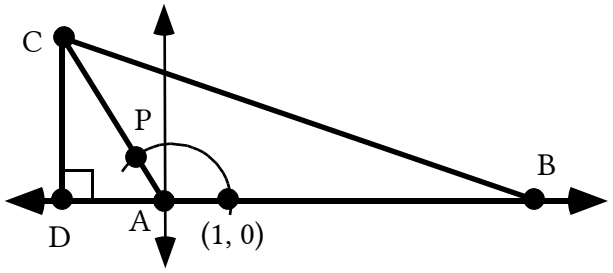

(0, 0, 2)
z
D
B
C
y
(1, 0, 0)
(0, 3, 0)
x
A

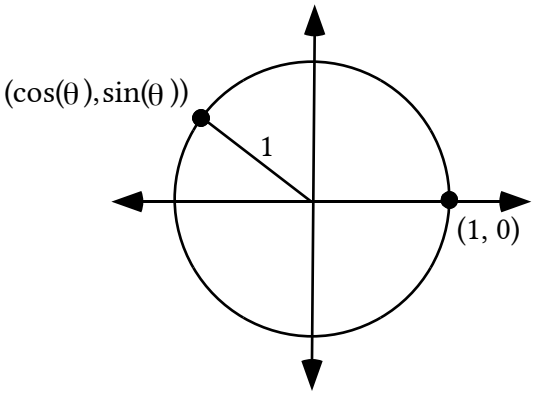

≈(3.656,213.891)
parabolic graph of
the solutions of
y=¯16 ·ξ2 +117 ·ξ
≈(0.998,100)
λινε γραπη οφ τηε
σολυτιονσ οφ
ψ=100
≈(6.324,100)


N
a
N
b
d
c

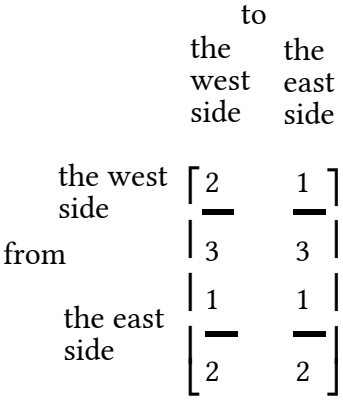

C
P
A
(1, 0)
B

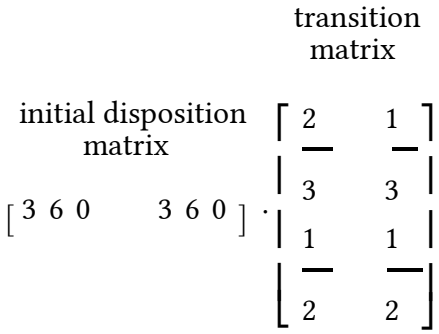

C
P
A
D
B

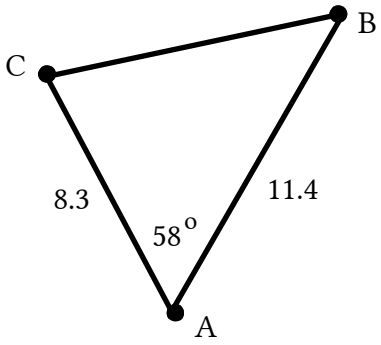

C
P
B
D
A
(1, 0)


(cos(θ), sin(θ))
1
(1, 0)

number on west side	number on east side
[3 6 0	3 6 0]


to
the west side
the east side
the west side
from
the east side
2
1
3
3
1
1
2
2


transition
matrix
initial disposition
matrix
3 6 0
3 6 0
2
3
1
3
1
2
1
2


B
C
8.3
11.4
58°
A